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Liquidation Mechanisms in Perp DEXes

GMX · dYdX v4 · Hyperliquid

A comparative study of design, safety, and the UX trade-off

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Topic J · May 2026

Why this matters

Perp DEXes settle **tens of billions** in daily notional. The liquidation engine decides:

- Who pays for losing trades
- How fast bad debt is closed
- **Who eats the loss when markets gap**

🚀

UX

high leverage, low fees, partial liquidations

Safety

conservative MM, deep insurance, fast oracles

Decentralization

permissionless keepers, no admin pause

No platform wins all three. This talk maps the trade-offs.

Vocabulary in 30 seconds

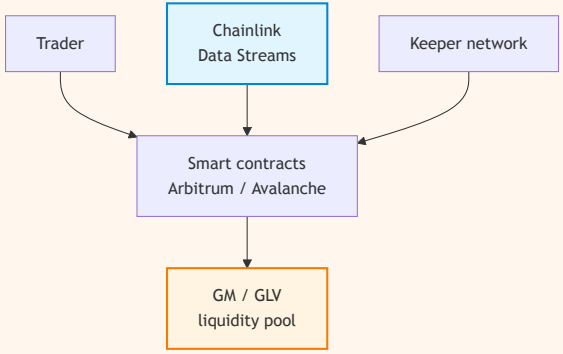
Term	What it means
Mark price	Price used for PnL & liquidation checks (≠ last trade)
IM / MM	Initial / Maintenance margin
Liquidator	Actor that submits the closing tx for a reward
Insurance fund	Pooled capital that absorbs shortfalls
ADL	Auto-deleveraging — force-close profitable positions to clear bad debt

Part 1

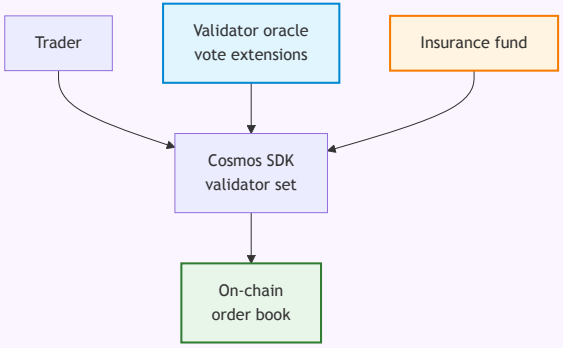
The Architectural Landscape

Three architectures, three philosophies

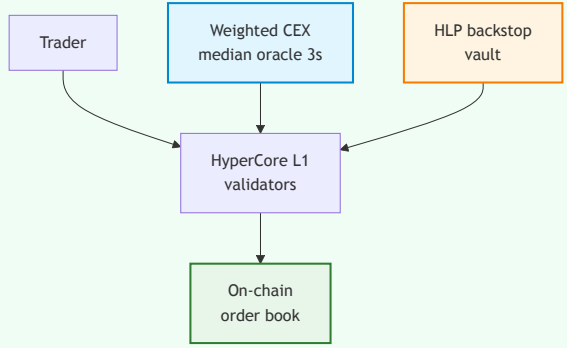
🟠 GMX v2 · Pool-counterparty



🟣 dYdX v4 · App-chain CLOB



🟢 Hyperliquid · CLOB + Vault



Architecture cheat-sheet

	GMX v2	dYdX v4	Hyperliquid
Venue	Arbitrum / Avalanche L2	Cosmos SDK app-chain	HyperCore L1
Matching	**No book** — AMM vs pool	On-chain CLOB	On-chain CLOB + HLP vault
Counterparty	GM / GLV LPs	Other side + insurance fund	Other side + HLP
Oracle	Chainlink Data Streams	Validator vote-extensions	Weighted CEX median, 3 s
Liquidator	Permissionless keeper bots	Liquidation daemon (any node)	Anyone (market) + HLP (backstop)

Leverage and margin

	GMX	dYdX v4	Hyperliquid
Max leverage	up to 100×	1–50× (tier-based)	3–40× (asset-tiered)
Modes	Pool collateral	Cross + isolated	Cross + isolated
MM rule	0.25–1 % floor	Fixed MMF per market	$\mathrm{MMR} = \mathrm{IMR}/2$, by tier

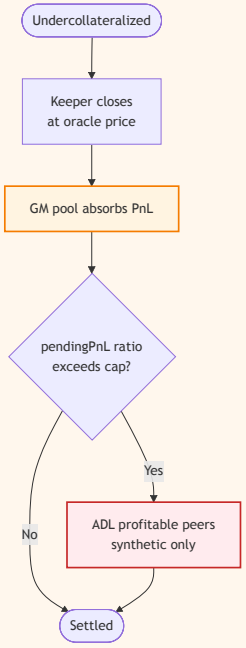
🔍 **Tier deep-cut (HL):** BTC 40× → drops to 20× above $150M OI per account.
Aggressive UX, but structural defense against whale cascades.

Part 2

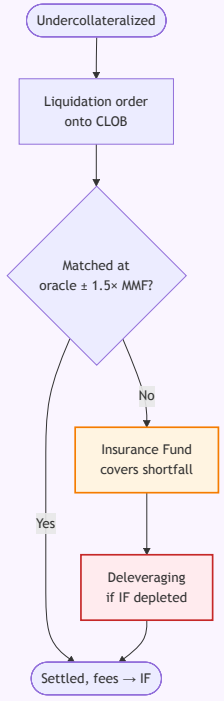
The Liquidation Waterfalls

Liquidation waterfall — side by side

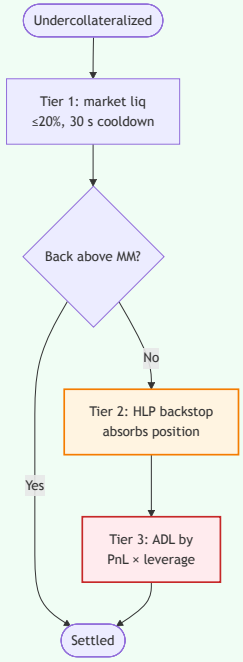
GMX



dYdX v4



Hyperliquid



👁 **Hyperliquid's red path:** Tier 1 Market Liq → 30 s cooldown → Tier 2 HLP backstop → Tier 3 ADL

Liquidation price formulas

GMX

Close at oracle adverse side:

- longs fill at `maxPrice`
- shorts fill at `minPrice`

Fee: **0.20 %** standard / 0.45 % synthetic

dYdX v4

Fillable price spread from oracle (sign s):

$$P_{\text{fill}} = P\big(1 - s \cdot \text{SMMR} \cdot \text{MMF} \cdot \text{BA} \cdot (1 - Q)\big)$$

$s = +1$ long-liq · $s = -1$ short-liq

Penalty up to **1.5 %** of account value → IF

Hyperliquid

On-chain liq price:

$$P_{\text{liq}} = P_{\text{mark}} - s\,\frac{m_{\text{avail}}}{|S|}\,\frac{1}{1 - l/l_{MM}}$$

No fee; MM buffer forfeited if HLP takes the position

Oracles: the trust delegation

Property	GMX	dYdX v4	Hyperliquid
Source	Chainlink Data Streams	Validator side-cars	Weighted CEX median
CEX list	Provider-internal	7+ exchanges	Binance 3×, OKX 2×, …
Cadence	Sub-second (pull)	~1 s (per block)	**3 s**
Aggregation	Provider-internal	Validator vote weight	Stake-weighted median
Defense	Two-phase commit	Governance + quorum	OI cap + 1 % resting-order cap

📌 Common theme: **avoid thin internal AMMs as oracle**. Lesson learned from 2021–2022 spot-AMM exploits.

Extreme-volatility defenses

	GMX	dYdX v4	Hyperliquid
Circuit breakers	Price-impact 0.5–10 %	None explicit	OI caps + 1 % resting-order cap
Delay auction	2-phase commit (~1 block)	None	3-s oracle + 30-s liq cooldown
Backstop	GM pool (`MAX_PNL_FACTOR`)	Insurance fund	HLP vault
ADL	Synthetic markets only	After IF exhausted	After HLP exhausted

Funding rate at a glance

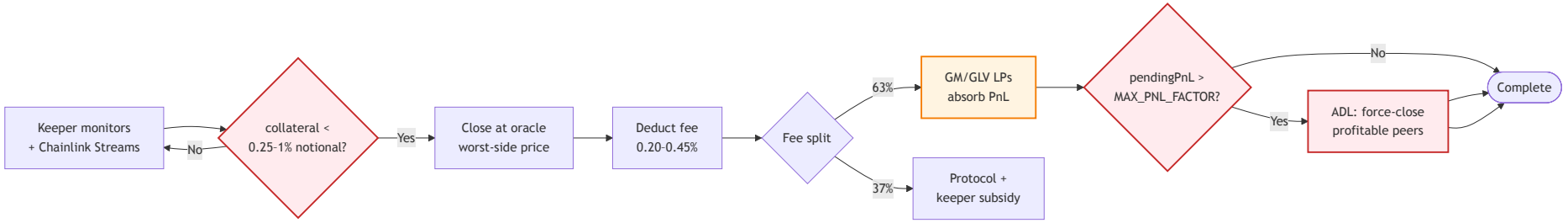
	GMX	dYdX v4	Hyperliquid
Type	Adaptive (OI imbalance)	Premium + interest	Premium + interest
Sample	per-second continuous	1 min	**5 s**
Settle	on position touch	hourly	hourly (1/8 of 8h rate)
Cap	15–300 % APR	$600\% \cdot (\mathrm{IMF} - \mathrm{MMF})$/8h	4 % per hour

Part 3

GMX Deep-Dive

The liquidation flow that backs $1B+ in TVL

GMX v2 liquidation flow



🔍 **Spotlight:** Keeper monitors → commit oracle (block N) → execute (block N+1) → **63%** fee → GM LPs / **37%** → protocol → ADL guard

Worked example — $1M long ETH @ 10×

Setup

- Entry: **$3,000**
- Drop to: **$2,730** (−9 %)
- PnL: **−$90,000**
- Collateral left ≈ $10,000 *(below 1 % floor)*

Liquidation

- Keeper triggers liquidation
- Fee: 0.20 % × $1M = **$2,000**
  - $1,260 → GM pool LPs (63 %)
  - $740 → protocol (37 %)
- Pool absorbs: **+$90,000** PnL (in WETH)
- Residual to trader: ≈ **$8,000**

> 💡 **GM pool ≈ short-vol, long-fee**
>
> Profits on liquidations; loses on sustained directional moves (capped by `MAX_PNL_FACTOR` → ADL).

GMX v1 → v2 evolution

Aspect	GLP v1	GM / GLV v2
Pool	Single shared basket	Per-market isolated pairs
Risk	One pool, all assets	Contagion contained per market
Oracle	Chainlink push feeds	Chainlink **Data Streams** (signed pull)
ADL	Limited	`MAX_PNL_FACTOR_FOR_ADL` per market

Lesson behind the upgrade: Sep 2022 AVAX manipulation on v1 GLP.

Part 4

Lessons from the Battlefield

Real incidents → real design changes

Date	Incident	Mechanism that failed	Modern fix
Mar 2020	"Black Thursday"	Maker 0-bid liquidations	Backstop bidders, min bid floors
May 2022	LUNA / UST	Reflexive collateral spiral	OI / asset caps
Sep 2022	GMX v1 AVAX	Thin spot + single pool	Per-market isolation
Nov 2022	FTX collapse	Centralized custody	Validated non-custodial perp DEX thesis
Mar 2025	**Hyperliquid JELLY**	HLP took toxic inventory	Asset onboarding + per-asset risk budgets

JELLY squeeze — the modern case

What happened

- Attacker self-traded a low-cap asset into liquidatable state
- HLP backstop absorbed the position *(as designed)*
- Underlying spot too illiquid for HLP to unwind
- HLP took a multi-million $ loss

What it teaches

Insurance-fund primitives need:

- Asset onboarding due diligence
- Per-asset risk budgets
- Governance kill-switch

> *"Automated backstop" isn't automatic enough for adversarial new listings.*

JELLY — quantitative attack model

Microstructure inequality *(necessary condition)*

$$\begin{cases} l \cdot c > L_{\text{exit}} & \text{book can't absorb} \\ l \cdot c \lesssim V & \text{HLP rich enough to swallow} \end{cases}$$

- For **BTC**: $L_{\text{exit}} \gg V$ → attack impossible
- For a **new listing**: inequality flips → HLP becomes the dump

Expected extraction *(dual-account model)*

$$\mathbb{E}[\text{extraction}] \approx \min\left(N\frac{p_1 - p_0}{p_0},\ V\right) - c$$

Plugging JELLY numbers ($N \approx \$70\text{M}$, $4\times$ ramp, $V \approx \$240\text{M}$, $c \approx \$3.5\text{M}$):

$$\approx \min(\$210\text{M},\ \$240\text{M}) - \$3.5\text{M} \approx \boxed{\$206\text{M}}$$

Profit-side addresses exit clean USDC *before* HLP can unwind — the liquidity time-lag is the rent.

Part 5

Open Problems & The Road Ahead

Open risks (common to all three)

1. **Oracle compromise** — Chainlink integrity vs validator collusion

2. **MEV / front-running** of liquidations — reduced but not eliminated

3. **ADL fairness** — being force-closed in profit is brutal UX

4. **Chain halt** — sovereign chains pause trading *and* liquidation

5. **LP / HLP impermanent risk** — short-vol decays in trends

6. **Stablecoin / bridge risk** — USDC depeg affects all three

7. **Cross-margin correlation** — MM ignores off-diagonal ρ_{ij}

8. **Funding-induced cascade** — capped funding drains margin even on flat mark

Cross-margin: the hidden correlation tax

Today's maintenance check *(diagonal only)*:

$$\mathrm{MM}_{\mathrm{req}} = \sum_i |S_i|\, P_i\, \mathrm{MMF}_i$$

But equity variance is $n \times n$:

$$\mathrm{Var}(E) = \sum_{i,j} S_i S_j\, \sigma_i \sigma_j\, \rho_{ij}$$

In crashes $\rho_{ij} : 0.5 \to > 0.9$ — ten alt-coin longs $\neq$ ten independent risks.

Why no on-chain fix yet? Cost.

Approach	Per-account cost
Full SPAN matrix H	$O(n^2)$ / block
Validator amortised	$O(n^3)$
Correlation-group haircut	$O(n + K)$

FTX / CME-SPAN-style: bucket assets into $K \ll n$ classes, assume $\rho=1$ inside, step-wise add-on on dominant net exposure.

Where the design space is going

Each idea pays a system-engineering price (in **red**).

🔐 **ZK-proven oracles** (Pyth Lazer, zkTLS) · cost: 200 ms–2 s prover latency → increases effective oracle staleness on 3 s-cadence chains

🏛️ **Intent-based liquidation auctions** → MEV value back to user · cost: window τ → bad debt $\Delta D \sim |S|\,v\,\tau$; linear in crashes

🤝 **Hybrid CLOB + RFQ** for liquidated inventory · cost: RFQ round-trip + mandatory CLOB fallback path = 2× engineering surface

🪨 **Per-asset insurance tranches** — no monolithic HLP · cost: fragmented capital → thin-asset vaults demand higher yields

✅ **Formal verification** of margin invariants · cost: protocol upgrades require re-proving; months per change

⚡ **Programmable circuit breakers** — per-asset LULD-style bands · cost: false-positive freezes legitimate hedging; vol-detector becomes governance risk

Three answers to one question

GMX

Isolate risk on willing LPs, trust Chainlink.

dYdX v4

Rebuild CEX inside a sovereign chain, transparent IF.

Hyperliquid

Aggressive leverage backed by a community vault.

None dominates. The next generation will **mix primitives** — oracles from one, insurance from another, matching from a third — to *break* rather than *navigate* the UX-vs-safety trade-off.

Questions?

Thank you

Report · Slides · Script — all in the project repo

Citation map: refs.md